CMPE 220 System Software: CPU Design

Professor: Ishie Eswar

Team Members (Group 6):

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GitHub Repository

GitHub Repository Link: https://github.com/ParthPatel00/CPU-Design

Git Clone Link: https://github.com/ParthPatel00/CPU-Design.git

How to Download, Compile, and Run

Requirements:

- macOS or Linux with GCC installed

- Git

- Make

Step 1: Download

git clone https://github.com/ParthPatel00/CPU-Design.git

cd CPU-Design/src

Step 2: Build

make clean && make

Cleans and rebuilds everything. Creates two executables: cpu (the emulator) and asm (the assembler).

Step 3: Assemble and Run Programs

To run the timer program:

./asm ../programs/timer.asm ../programs/timer.bin

./cpu run ../programs/timer.bin

To run the hello world program:

./asm ../programs/hello.asm ../programs/hello.bin

./cpu run ../programs/hello.bin

To run the fibonacci program:

./asm ../programs/fibonacci.asm ../programs/fibonacci.bin

./cpu run ../programs/fibonacci.bin

Expected output:

- hello: Hello, World

- fibonacci: 0 1 1 2 3 5 8 13 21 34

- timer: [TIMER] 3 ... [TIMER] 1

- multiply: 6 (recursive 3 * 2)

Step 4: Verbose Mode (run one at a time)

./cpu verbose ../programs/hello.bin

./cpu verbose ../programs/fibonacci.bin

./cpu verbose ../programs/timer.bin

./cpu verbose ../programs/multiply.bin

Shows every Fetch / Decode / Execute cycle in full detail.

Team Member Contributions

Parth Patel:

Designed overall CPU architecture, instruction set, instruction encoding, and memory map. Implemented emulator modules: cpu.c (register file), alu.c (arithmetic and flag logic), control.c (fetch/decode/execute loop), bus.c (memory routing), memory.c (64K word array and binary loader), io.c (memory-mapped IO devices), and main.c (CLI entry point). Created CPU architecture documentation and schematic.

Harshitha Vadavalli:

Implemented two-pass assembler (assembler.c, asm_main.c). Wrote all four assembly programs: timer.asm, hello.asm, fibonacci.asm, and multiply.asm. Created Makefile build targets.